THE COLIMIT OF HOM COMPLEXES OUT OF THE KOSZUL COMPLEXES IDENTIFIES WITH THE TOTAL ALTERNATING CECH COMPLEX OF A COMPLEX OF MODULES

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Allowed standard prerequisites (editorial).

- The alternating Cech complex of a complex of modules is the total complex of the double complex of alternating Cech groups, with conventions recorded in the cohomology Cech section. invoked at proof lines 2246-2250. Fixes the right-hand double complex F and its differentials.
- The maps psi from the Hom colimit double complex to the Cech double complex constructed in the single-complex alternating lemma are termwise isomorphisms compatible with the first differential. invoked at proof lines 2257-2259. Reduces the comparison of the two double complexes to a sign fix.
- The hom complex of complexes of modules carries its own differential convention, recorded in the More on Algebra hom-complex section. invoked at proof lines 2262-2265. Supplies the left-hand differential whose sign differs from the total complex convention.

Source reference. Original remark 08CZ: sources/stacks-chow/tags.txt, source lines 10619–10619. Cited at statement line 2231 as the standing setup; the double complex comparison is inside the counted span.

Source reference. Original section 01FP: sources/stacks-chow/tags.txt, source lines 1878–1878. Cited at proof lines 2252-2253 for the right-hand double complex conventions; the comparison is inside the counted span.

Source reference. Original lemma 08D0: sources/stacks-perfect/pinned-original-source.tex, source lines 2128–2128. Cited at proof lines 2258-2259 for the termwise isomorphisms psi; the double complex upgrade is inside the counted span.

Source reference. Original section 0A8H: sources/stacks-chow/tags.txt, source lines 12915–12915. Cited at proof lines 2264-2265 for the left-hand differential; the sign correction is inside the counted span.

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Lemma 0.2. *In Situation 08CZ. Let $M^\bullet$ be a complex of A-modules and denote $\mathcal{F}^\bullet$ the associated complex of $\mathcal{O}_X$-modules. Then there is a canonical isomorphism of complexes*

$$\operatorname{colim}_e \operatorname{Hom}^\bullet(I^\bullet(f_1^e, \ldots, f_r^e), M^\bullet) \longrightarrow Tot(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet))$$

functorial in $M^\bullet$.

Proof. Consider the double complex $F^{\bullet,\bullet}$ with terms $F^{p,q} = \mathcal{C}^p_{alt}(\mathcal{U}, \mathcal{F}^q)$ discussed in Cohomology, Section 01FP. Consider the double complex $G^{\bullet,\bullet}$ with terms $G^{p,q} = \operatorname{colim}_e \operatorname{Hom}_A(I^{-p}(f_1^e, \ldots, f_r^e), M^q)$ and differentials given by functoriality (without the intervention of signs). The maps $\psi^{p,q} : G^{p,q} \to F^{p,q}$ constructed in the proof of Lemma 08D0 are isomorphisms and compatible with the differentials d_1 (by the lemma) and d_2 (this is clear). However, the differentials d on the complexes on the left and right hand side of the arrow in the lemma have different signs. Namely, for $g \in G^{p,q}$ is given by

$$d(g) = d_2(g) - (-1)^{p+q} d_1(g)$$

(see More on Algebra, Section 0A8H) and the differential for $f \in F^{p,q}$ is given by

$$d(f) = d_1(f) + (-1)^p d_2(f)$$

Thus we can fix the signs by multiplying $\psi^{p,q}$ by $(-1)^{pq+p(p-1)/2}$. □